Week 9: Iteration Log

draft notes

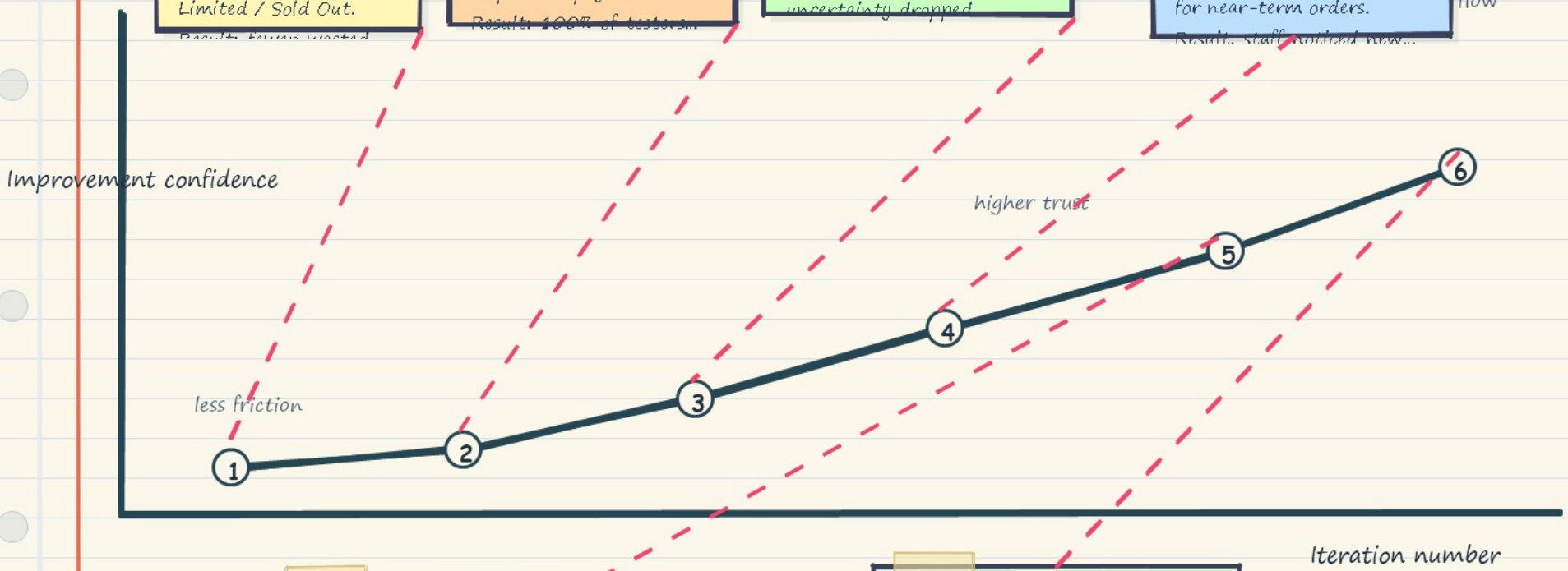
Qualitative impact chart from the reported build - test - feedback - refine cycle

1. Menu availability
Badges show Available / Limited / Sold Out. Result: fewer wasted

2. Time-slot picker
Moved to a mandatory step before payment. Result: 100% of testers

3. Payment feedback
Spinner + success state after UPI scan. Result: payment uncertainty dropped

4. Dashboard alert
Chime + amber highlight for near-term orders. Result: staff noticed new...



5. Staff earnings tab
Revenue, order count, and item-level sales. Result: better daily monitoring.

6. Checkout summary
One clear page and one place-order action. Result: completion rate improved.

Interpretation note: the trend line is qualitative and derived from the report outcomes, not from numeric benchmark scores.

Week 10: Ethics and Inclusivity

ethics check

Hand-drawn risk-to-safeguard chart for a responsible college canteen platform

Risks Raised in the Report

Fraud or fake buyers
False orders can create waste and operational chaos.

Data leakage
Phone numbers, UPI details, and wallet history need protection.

Digital exclusion
Low-tech or visually challenged students could be left out.

Job displacement fear
Staff may worry that automation replaces their role.

Safeguards and Inclusive Design

Confirmed payment first
UPI confirmation before processing blocks no-show orders.

Security controls
JWT expiry, bcrypt hashing, encryption, and limited retention.

Accessibility choices
Basic Android support, text scaling, contrast, language options.

Workflow enhancement
Staff still prepare, serve, and manage daily operations.

Positive impact: more trust, less stress, fairer access, and calmer work conditions.
Stakeholders named in the report: students, canteen staff, faculty, administration, and management.

Week 11: Digital Prototyping

stack sketch

Hand-drawn three-tier stack chart for the working SRM Kitchen prototype

working prototype, not just static screens

Tier 1: Frontend Experience Layer

Student Portal
React.js + Vite
Tailwind CSS

App Flow
Redux Toolkit state
React Router v6

Token + Alerts
QRCode.js pickup token
React Toastify notices

Role Views
student / staff / admin routes
mobile-first layout

API calls, auth, status sync

Tier 2: Backend and Realtime Logic

Node.js + Express.js APIs
REST endpoints for auth, menu, orders, wallet

JWT + bcryptjs security
role-based access and password hashing

Socket.IO updates
live order status between students and kitchen

persistence, menu data, transactions

Tier 3: Database and Domain Models

MongoDB Atlas + Mongoose
cloud database, schema validation, ODM layer

Core models
User, MenuItem, Order, Wallet, Feedback

Order workflow
pending -> preparing -> ready -> collected / cancelled

Development tools
GitHub, MongoDB Compass, Postman, Figma, VS Code + ESLint

System view from the report
Student Portal + Kitchen Hub + Admin Oversight in one synchronized working product